4. Limitations

This section lists the main limitations of the data, the targets, the models and the app. They should be read together with the results in Sections 2 and 3. Several results that look strong (such as $R^2 = 1.0$) are only strong because of these limitations.

4.1 The dataset looks synthetic

Several patterns in the data do not match what is normally seen in real estate markets:

- **Flat distributions:** Price, size and BHK are almost evenly spread across their ranges. Real housing prices are usually skewed, with more properties in the lower and middle price bands.
- **No price-size relationship:** Larger properties do not cost more. The scatter plot of size against price shows no trend (Section 2, Q4).
- **Identical city prices:** The average price of the top 15 cities differs by less than ₹7 lakh. Mumbai and New Delhi, which are among the most expensive cities in India, appear at the bottom of the list (Section 2, Q7).
- **Even BHK split:** Each BHK type makes up about 20% of properties in every city. Real cities have many more 2 and 3 BHK homes than 5 BHK homes (Section 2, Q8).
- **Small discrete features:** Nearby_Schools only takes values 1 to 10, Nearby_Hospitals 1 to 10, and Amenities_Count 1 to 5.
- **Equal category counts:** Owner types, availability statuses and parking are almost exactly split in half or into thirds.

Impact: Findings from this dataset describe the dataset, not the real Indian housing market. Conclusions such as "parking does not affect price" or "schools do not affect price" should not be taken as real-world facts.

4.2 The Good Investment label is a rule, not an outcome

- The label is defined as "price per sq ft below the median of the city". It is not based on rental yield, resale value, or what buyers actually earned.
- Because the label is defined this way, about 50% of properties are "Good" by construction.
- The classifier is learning to reproduce this rule from other features. Its accuracy (about 71%) reflects how well those features predict the rule, not how well they predict investment success.

4.3 The 5-year future price is a formula, not a forecast

- The dataset does not contain real future prices. The 5-year price was calculated with a compound growth formula.
- The city growth rates (for example, Bangalore 10%, Kolkata 6%) are assumptions made for this project, not measured data.

- Real property prices depend on many factors that are not in the model, including interest rates, inflation, local supply and demand, government policy and economic cycles.
- The R^2 value of 1.0 reflects how well the models reproduce the formula. It does not show accuracy on real future prices.

Impact: The regression models should not be used to predict actual future prices. They are a demonstration of how the pipeline works, and the growth rates could be replaced with real historical data if it becomes available.

4.4 The classifier depends mainly on size

- Size_in_SqFt accounts for about 68% of the classifier's feature importance (Section 3.1).
- In this dataset, price per sq ft falls as size rises. The model therefore learned "larger properties are more likely to be Good Investments."
- This is a pattern in the data, not a sound investment principle. In real markets, larger homes often have a lower price per sq ft, but that does not make them better investments.

Impact: The app should not be used to recommend larger properties as better investments. A proper test would remove Size_in_SqFt from the classifier and check the accuracy. This test is recommended for future work.

4.5 Missing features

The following features were mentioned in the project brief but are not available in the dataset:

Feature in brief	Status	How it was handled
Crime rate	Not in dataset	Not used. Security (Yes/No) used as a rough proxy
Infrastructure score	Not in dataset	Not used. Public transport access and nearby schools and hospitals used as proxies
School density score	Not in dataset	Not created (needs area data)
Sale date	Not in dataset	Time trends cannot be measured (Section 2, Q10)
Locality coordinates	Not in dataset	Location analysis uses locality names only
Real future prices	Not in dataset	Future price is calculated from a formula
Rental income	Not in dataset	Investment return is not measured

4.6 Category mismatches with the brief

- **Owner types:** The brief lists Individual, Builder and Agent. The dataset uses Owner, Builder and Broker.
- **Availability status:** The brief lists Available, Under Construction and Sold. The dataset only has Ready_to_Move and Under_Construction.
- **Parking:** The brief describes the number of parking spots. The dataset only has Yes or No.

These differences were handled by using the dataset values. The brief's questions about these features can only be answered for the categories that exist in the data.

4.7 Model selection and evaluation

- **No separate validation set:** The best model was selected using the test set. This means the reported test scores are slightly optimistic, because the choice of model was influenced by the same data used to report performance. A three-way split (training, validation, test) would avoid this.
- **Sampling for Random Forest:** Random Forest was trained on a random 60,000-row sample of the 200,000 training rows, to keep training time reasonable. This may slightly reduce its accuracy compared with training on all rows.

- **Small differences between classifiers:** The three classifiers differ by less than 0.5 percentage points in accuracy. The choice between them is not strongly supported by the results.
- **Fixed decision threshold:** The classifier uses a probability of 0.5 as the cutoff for "Good". This cutoff was not tuned, and a different cutoff might give different precision and recall.
- **Overfitting check:** The comparison of training and test scores is recorded in Section 3.4. If the gap is large, the model may be memorising the training data rather than learning general patterns.

4.8 Data cleaning effects

- **Capping outliers:** Price and size were capped at the 1st and 99th percentile. This removes the most extreme values and makes the distributions look more uniform than the raw data. The capping may hide real high-end and low-end markets.
- **Price per sq ft outliers kept:** About 7.9% of properties have unusually high price per sq ft values (Section 2, Q5). These were kept in the data, which means averages are pulled upward by these values. Medians are used where possible for this reason.
- **Growth rate defaults:** Only 8 cities have a specific growth rate. All other cities use the default rate of 8%, which means most properties in the dataset use the same rate.

4.9 Deployment and app limitations

- **Sample data in the app:** The Streamlit app uses a random 30,000-row sample of the data for the price comparison and the Explore tab. The models themselves were trained on all 250,000 rows, but the app's peer comparison is based on the sample only.
- **Library version pinning:** The app's requirements are pinned to the versions used in training (scikit-learn 1.6.1, XGBoost 3.4.1, pandas 2.2.3). If the environment changes, saved models may not load correctly.
- **No live data:** The app does not connect to any live property listing or market data source.
- **No user feedback:** The app does not record user inputs or check whether predictions turned out to be correct.

4.10 Scope of use

This project is an academic demonstration of a data science pipeline. It should not be used as:

- Financial advice, or a basis for buying or selling property
- A valuation tool for real transactions
- A replacement for professional valuation by a licensed property valuer
- A source of decisions about specific neighbourhoods, cities or groups of people

Any real-world use would require real transaction data, validation against actual outcomes, and review by domain experts.